

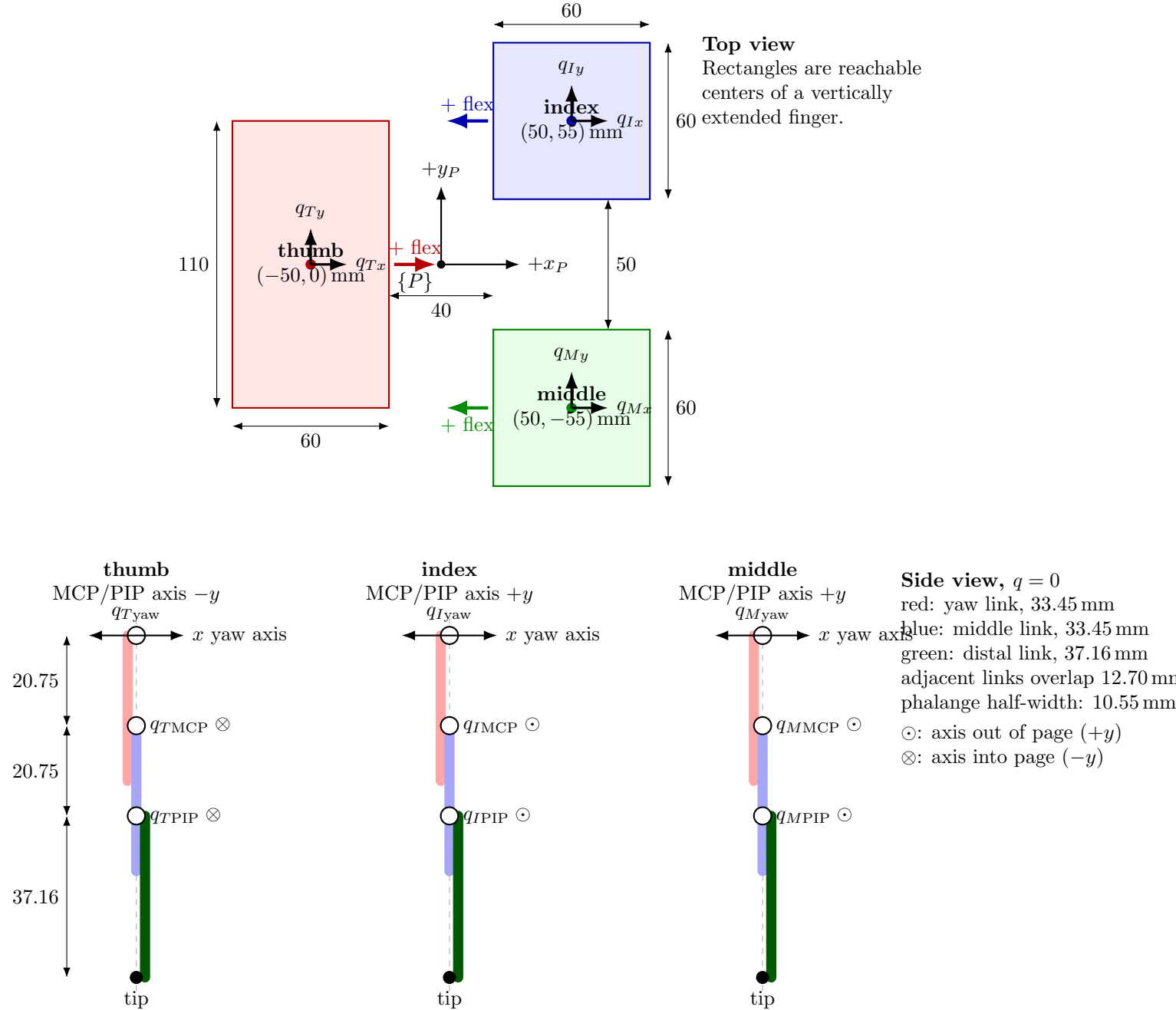
15-DoF Reconfigurable Three-Finger Hand: Kinematic Layout

Six XY morphology DoFs + nine finger articulation DoFs. The external 6-DoF palm pose used in the MuJoCo scene is not counted.

The CAD workspace drawing is registered to the MuJoCo palm frame by centering the overall CAD envelope at (55, 80) mm and rotating it so that the thumb opposes the other fingers along the MuJoCo x direction:

$$x_{MJ} = -(y_{CAD} - 80 \text{ mm}), \quad y_{MJ} = -(x_{CAD} - 55 \text{ mm}).$$

Thus zero morphology places each finger at the center of its rectangular XY workspace.



Morphology workspace parameters

Finger	x_0 [mm]	y_0 [mm]	q_x range [mm]	q_y range [mm]	full workspace [mm]
Thumb	-50	0	$[-30, 30]$	$[-55, 55]$	60×110
Index	50	55	$[-30, 30]$	$[-30, 30]$	60×60
Middle	50	-55	$[-30, 30]$	$[-30, 30]$	60×60

Complete 15-DoF internal-joint parameter table

For each finger $f \in \{T, I, M\}$, the first two joints translate the entire finger/yaw frame in the palm plane. The remaining three joints form the articulated finger chain. Axis directions below are given at the zero configuration in the palm frame.

Finger	DoF	Type / axis	Range	Home-frame location
Thumb	q_{Tx}	P, $+x_P$	$[-30, 30]$ mm	$(-50, 0, 0)$ mm
	q_{Ty}	P, $+y_P$	$[-55, 55]$ mm	
	q_{Tyaw}	R, $+x$	$[-85, 85]^\circ$	yaw center
	q_{TMCP}	R, $-y$	$[-15, 92]^\circ$	20.75 mm below yaw
	q_{TPIP}	R, $-y$	$[-18, 92]^\circ$	41.50 mm below yaw
Index	q_{Ix}	P, $+x_P$	$[-30, 30]$ mm	$(50, 55, 0)$ mm
	q_{Iy}	P, $+y_P$	$[-30, 30]$ mm	
	q_{Iyaw}	R, $+x$	$[-85, 85]^\circ$	yaw center
	q_{IMCP}	R, $+y$	$[-15, 92]^\circ$	20.75 mm below yaw
	q_{IPIP}	R, $+y$	$[-18, 92]^\circ$	41.50 mm below yaw
Middle	q_{Mx}	P, $+x_P$	$[-30, 30]$ mm	$(50, -55, 0)$ mm
	q_{My}	P, $+y_P$	$[-30, 30]$ mm	
	q_{Myaw}	R, $+x$	$[-85, 85]^\circ$	yaw center
	q_{MMCP}	R, $+y$	$[-15, 92]^\circ$	20.75 mm below yaw
	q_{MPIP}	R, $+y$	$[-18, 92]^\circ$	41.50 mm below yaw

DH representation of each 3R finger subchain

A classical DH table is most useful for the 3R articulated subchain; the two orthogonal base slides are represented directly as prismatic joints in the table above. One consistent standard-DH frame assignment is:

i	Joint	θ_i	d_i	a_i	α_i
1	yaw	q_{fyaw}	0	20.75 mm	$\sigma_f 90^\circ$
2	MCP	q_{fMCP}	0	20.75 mm	0°
3	PIP	q_{fPIP}	0	37.16 mm	0°

where $\sigma_T = +1$ and $\sigma_I = \sigma_M = -1$ for this frame choice. The sign change reflects the reversed thumb flexion-axis convention. DH frame signs are not unique; the MuJoCo axis vectors in the 15-DoF table above are the authoritative implementation convention.

Physical-link geometry. The yaw and middle physical links are each 33.45 mm from rotational-axis center to their distal surface. Because the next joint center occurs 20.75 mm downstream, each pair overlaps by $33.45 - 20.75 = 12.70$ mm. The distal link is 37.16 mm from PIP axis to fingertip surface. The capsule radius is 10.55 mm; consequently the MuJoCo capsule medial-axis endpoints are 22.90 mm, 22.90 mm, and 26.61 mm so that the rounded surfaces terminate at the measured CAD dimensions.